Algorithmic Intelligence in the Quantum-AI Era: Making Smart Decision for Complex Systems

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Society of Algorithmic Intelligence (SoAI)

This talk is based on collaborations with Ralf Borndörfer, Maria Grith, Thorsten Koch, Hannah Lai, Kailiang Liu, Jianlong Lu, Hanqiu Peng and Hongrui Zhang

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NUS National University of Singapore | Faculty of Science | Faculty of Arts & Social Sciences | RMI Advancing Risk Management for Singapore and Beyond | Institute of Operations Research and Analytics | Institute of Data Science | AI·DF Asian Institute of Digital Finance | IASC-ARS The Asian Regional Section of The International Association for Statistical Computing | SoAI SOCIETY OF ALGORITHMIC INTELLIGENCE

Complex Systems — where decisions depend on Algorithmic Intelligence

What is a complex system?

- Many interacting agents/variables with feedback loops and network effects
- Nonlinearity & emergence; heavy tails, regime shifts, rare events
- Constraints & multi-objective trade-offs (cost, risk, fairness, well-being)
- Combinatorial decisions (assignment/rostering/routing) under uncertainty and limited time

Why it matters now (urgency):

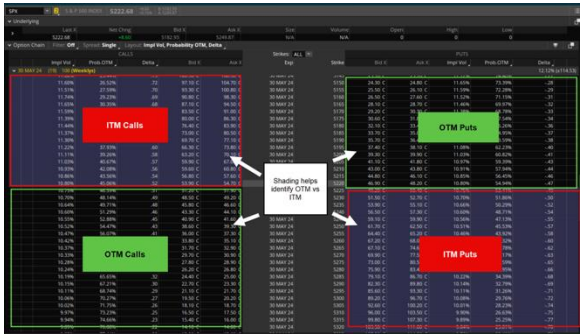
- Decisions increasingly automated; AI models drive actions in finance, healthcare, logistics, cyber
- Distribution shift & drift are pervasive; actions change the data we see (closed loop)
- Time-to-decision is critical; classical exact solvers often too slow at surge scale
- Governance expectations (reliability, fairness, auditability) are rising

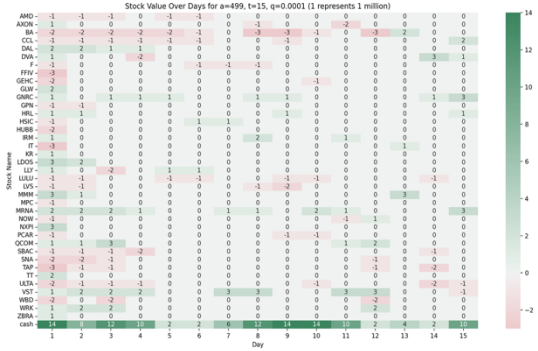
More and more, the decisions in these systems are being made by algorithms.

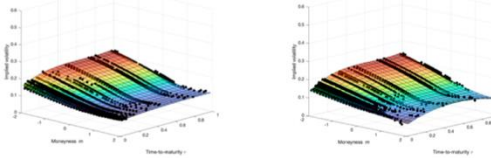
The Challenge of COMPLEXITY & SCALE (The "What")

The problems we face are no longer simple. They involve millions of interacting variables/agents.

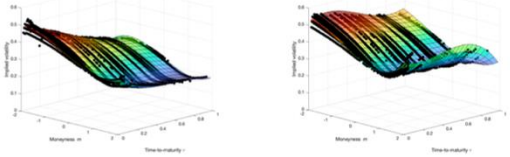
- Forecasting implied volatility (~6.4 million put and call SPX contracts from Jan 2009 to Dec 2021): You're not just forecasting a number; you're modeling a dynamic, high-dimensional surface.
- Multi-period portfolio trading (499 stocks over 2 weeks): This isn't a single trade; it's a complex sequence of decisions across a vast asset universe, considering transaction costs and market impact.







(a) Call, Apr 02, 2019 (b) Put, Apr 02, 2019



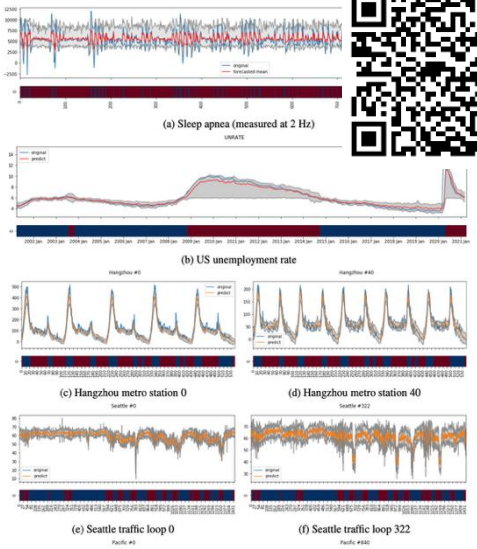
(c) Call, Apr 09, 2020 (d) Put, Apr 09, 2020

These are not toy problems; they are massive decision spaces. And no human can possibly manage them manually unaided.

The Challenge of UNCERTAINTY & DYNAMICS (The “Why It's Hard”)

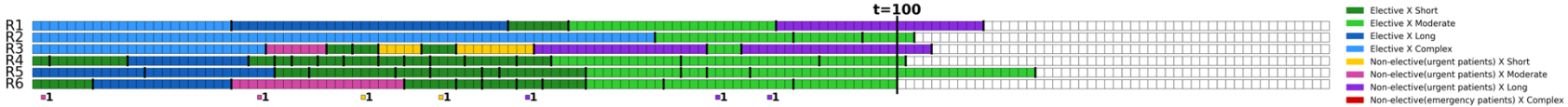
Complexity alone isn’t the full story. These systems are non-stationary and riddled with uncertainty.

- Hospital Operation Room (OR) Scheduling is a perfect storm of uncertainty: emergency arrivals spontaneously, surgery duration variability from expected, human factors (staff availability).
- Regime Switching: the uncertainty of switching between different regimes (e.g., calm/crisis over night, bull/bear markets, economic cycles, sleep apnea events).
- In logistics, ships are delayed, storms appear, regulations change.



The challenge isn’t just solving a problem for now — it’s ***adapting under uncertainty, and in real time.***

RL solution: EP 51/55-NEP 7/7-OT 33-CR 756.11



The Challenge of MULTI-OBJECTIVE OPTIMIZATION (The “What We Need”)

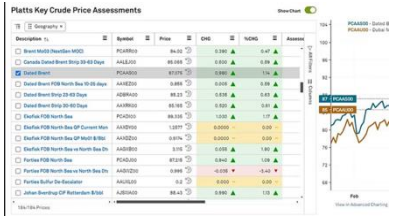
The goal is never singular. We must balance competing priorities in a principled way.

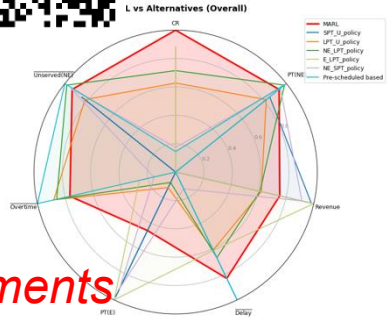
- Hospital OT Scheduling: You're not just maximizing revenue. You're balancing it with patient satisfaction and staff wellbeing—a fundamentally statistical trade-off.
- Bunker procurement (Fleet of 6 vessels, 42-day voyage with a large container shipping company over 2.5 years including high volatility periods e.g. Russia–Ukraine war): The goal is to minimize cost (saving millions) while managing the risk of price volatility and ensuring operational continuity.

Revenue & reputation

Quality of Patient Care

Well-being of medical staff





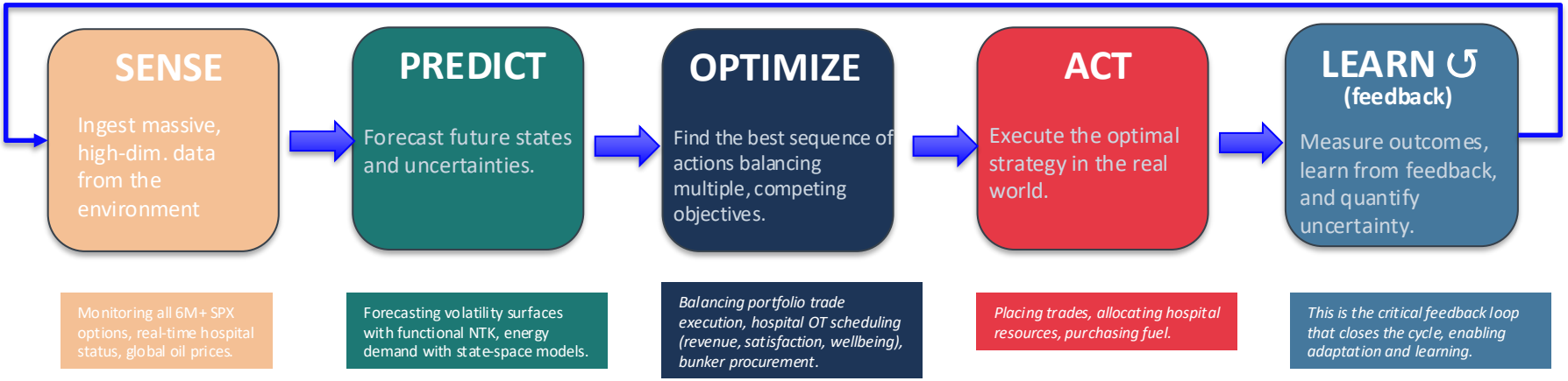
These trade-offs are inherently stochastic: we must quantify, balance, and monitor them.

A model without algorithmic intelligence is like a plane without instruments — it may fly fast, but it’s flying blind."

The Engine of Decision-Making: The Algorithmic Intelligence Cycle

"Human intuition alone cannot navigate the complexity of modern systems.
We need a disciplined, scalable cycle of Algorithmic Intelligence."

Algorithmic Intelligence integrates core STEM pillars—applied math, computer science, statistics & data science, optimization, and engineering—into a unified engine for action.



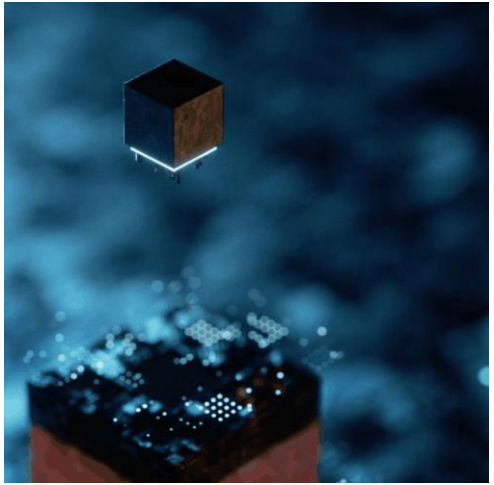
- ❑ What it is: a closed loop — sense → model → optimize → act → learn — delivering decisions (not just predictions), with targets (utility/risk), constraints (capacity/fairness), and in feedback with the real world.
- ❑ What it isn't: not just high predictive accuracy;

A smart model without calibration, optimization, and governance, ≠ smart decisions.

AI × QC are a "heaven match" for Complex Systems

The mathematical problems at the heart of this backbone—massive-scale optimization, HD inference, and stochastic modeling—are often intractable for classical computation.

- **Artificial Intelligence (AI)** provides the scalable function approximators (e.g., neural networks) and learning algorithms (e.g., reinforcement learning) that solve the core problems of sensing and prediction within this mathematical framework for the complex system.
- **Quantum Computing (QC)** adds a new gear to the compute stack. Paired with modern HPC, it is expected to accelerate the combinatorial and algebraic cores of decision making. Quantum computing is deeply connected to statistics — both aim to model uncertainty and explore complex probability spaces.
  - At its core, a quantum state is a probability distribution evolving through linear transformations, making quantum algorithms statistical by nature.
  - This connection opens opportunities for statisticians to contribute to quantum experiment design, error modeling, uncertainty quantification, and benchmark validation — ensuring that quantum results are not only faster, but also statistically sound and interpretable.



Together, AI and QC **do not** replace the mathematical backbone; they are however its most powerful accelerators: AI makes sense of the data; QC explores the combinatorial heart. When guided by statistics, they provide the necessary computational force to apply rigorous mathematical principles to the vast complexity of the real world.

AI engine is here, QC engine is scaling

Grants & national programs

U.S. National Quantum Initiative
$3.2B (2023–2025)

UK National Quantum Strategy
$3.5B (2023–2025)

DE National public investment
$3.2B (2023)

JP. National public investment
$7.4B introduced Jan 2025

EU Quantum Flagship
€1B over 10 years (2018–2028)

Singapore National Quantum Strategy
S$300 millions over 5 years (2024–2029)

Industry pilots & ecosystem

McKinsey's Quantum Report
revenue $650–750M (2024)
Startup funding +50% YoY to $2B (2024)
Market may hit $100B by 2040

IBM Quantum Network
≈300 orgs; 600k+ users (2025)
$150 Billion Investment in America to accelerate technology opportunity

Port of Los Angeles (SavantX × D-Wave)
Digital-twin yard ops optimized via QA (2024)
Crane productivity +60%; truck wait time −30%

HSBC × Quantinuum
Quantum-safe pilot for tokenised gold (2024)

Papers & research activity

Research Fundings
Global quantum funding: $44.5B

Papers
464k **AI** articles (1991-2020)
~50% published since 2016

Patents
AI 14,000+ (~38% CAGR, 2014-2023)*
Quantum issued +61% YoY (2023)**

The engines are here; Applied Math must now step into the driver's seat.

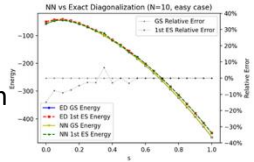
Global AI investment was $252 billion in 2024.
*WIPO-Patent Landscape Report (GenAI), AI patent grew from **733 (2014)** to **14,000+ (2023)**.
**QED-C report, issued quantum computing patents in 2023 were +61% vs 2022 (global).

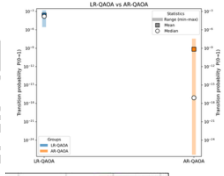
Quantum × AI: Intelligence Beyond Boundaries

Integrating Quantum Computing and Artificial Intelligence empowers scalable, accurate, effective, efficient, and sustainable solutions to tackle massive-scale optimization, multi-objective decision making, high-dimensional inference, and stochastic modeling beyond classical limits.

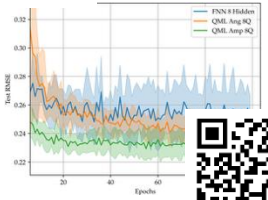
Quantum Optimization: Adaptive Ramp (AR)-QAOA

A quantum annealing–style schedule with nonlinear, FOAPT-guided β_k/γ_k ramps that suppress excitations and outperform LR-QAOA, benchmarked on IBM Eagle/Heron processors up to 156 qubits ($p \in [10, 100]$). Our optimization strategy for AR-QAOA beats the linear ramp QAOA with win rate 100% and improvement with many orders of magnitude.



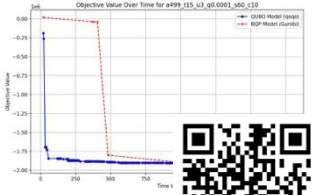


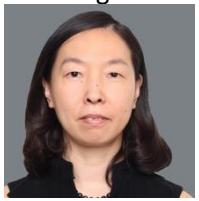
Quantum ML: Amplitude-encoded hybrid Quantum Neural Network for credit risk: Compresses 256 features into 8 qubits for recovery rate prediction with 1,725 samples only, achieving scalable, parameter-efficient learning ($O(3 \log_2 N)$) and significantly lower forecast errors than classical FNNs and angle-encoded QNNs in four-fold cross-validation.

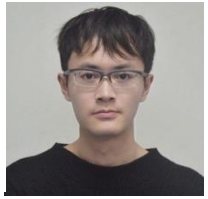


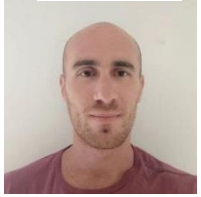
Quantum Optimization Benchmarking Library - The Intractable Decathlon: QOBLIB standardizes quantum-optimization benchmarking that links ten challenging problem classes to practically relevant applications. This shared platform allows the community to track — with fairness, transparency, and reproducibility — how close we are to achieving genuine quantum advantage in optimization.

Quantum Portfolio Optimization: Dynamic multi-objective trading strategy over 499 S&P 500 stocks (3 weeks) with transaction costs and short-selling; delivers faster, and accurate allocation—qoqo converges significantly quicker than Gurobi for near-optimal ($\pm\epsilon$) solutions.









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SOAI

SOCIETY OF ALGORITHMIC INTELLIGENCE

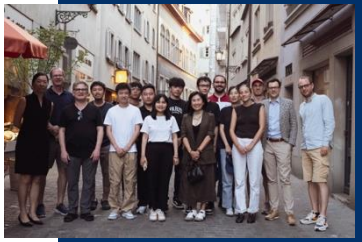
Leading Emerging Technology & Algorithms Research Society

Society of Algorithmic Intelligence

A global community dedicated to advancing the science and practice of intelligent algorithms. We bring together leading minds from academia, industry, and government to explore the frontiers of artificial intelligence, machine learning, and quantum technologies.

Become a Member →

Learn More



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We are organizing a 3-day Quantum×AI Conference on 24-26 and 28 September 2026 in Singapore

- **Keynotes** by leading researchers from academia and industry
- **Parallel sessions** on quantum computing and AI foundations and applications
- **Hands-on tutorials** on
  - Quantum Computing fundamentals, optimization, forecasting, and hybrid QxAI workflows
  - AI Algo Trading
  - AI agent for coding

  supported with user cases mainly in finance, transportation and healthcare. Public training sessions for practitioners and researchers on quantum computing and AI foundations and implementations.
- **Panel discussions and roundtables** fostering academic–industry collaboration, funding and VC opportunities in SG, Asia and Global
- **Ecosystem forum**: industry–academia showcases and collaboration opportunities.
- **An executive track (by invitation only)** designed for senior technology and innovation leaders, offering strategic briefings and curated roundtables for decision-makers

More: https://www.soc-ai.org/events/quantum-ai-risk-management

Global Certificate Program for Emerging Technologies

Emerging technologies — including Generative AI, Quantum Computing, and Decentralized Finance — are rapidly transforming industries and redefining competitive advantage before their theoretical foundations are fully settled. This creates a fundamental mismatch: Academia prioritizes rigor, validation, and stability vs Industry prioritizes speed, experimentation, and value creation

The Society of Algorithmic Intelligence (SoAI) introduces a new educational paradigm: globally integrated, research-driven, and application-focused certificate programs designed to bridge this gap. This positions SoAI as a first-of-its-kind global platform for the education of emerging technologies. Unlike traditional programs, SoAI combines:

Frontier research leadership, Hands-on, application-centric training, Multi-country, multi-institution delivery, Direct industry engagement

Since 2022, we have been organizing a series of **quantum computing events** focused on real-world applications and industry engagement.

- *Frontiers in FinTech and Quantum Computing* (2022),
- *Accelerating Industrial Innovation with Quantum Computing and Data Science* (2023),
- Computational Optimization at Work Co@Work (2024 in Berlin)
- *Quantum Computing Workshop – AI, Optimization, and Forecasting Across Industries: Digital and Quantum Computing* (2024),
- *ETH–NUS AI×FinTech Workshop and Hackathon* (2025)
- *ETH Zürich Fintech & Quantum Finance Workshop (2025 in Zurich)*
- *The 9th ISM-ISCT-NII-ZIB-NUS-MODAL Workshop on Optimization and Machine Learning for Data Science and Future Computing (2025 in Tokyo)*
- MODAL Berlin Algorithmic Intelligence Workshop 2025: Bridging AI, Quantum, and Optimization Empowering Industrial Innovation Through Real-World Problem Solving and Academic-Industry Collaboration (2025 in Berlin)

Each event has brought together **industry technology leaders and academic researchers**, combining hands-on tutorials, hackathons, and/or collaborative discussions to bridge research and practice.

Global Certificate Program for Emerging Technologies

Certificate in Generative AI: From Models to Deployment

This program focuses on the transition of generative AI from research to operational systems. Participants work with large language models, multimodal architectures, and AI agents to develop end-to-end solutions from model adaptation to deployment. Hands-on components include fine-tuning, prompt engineering, and building agent-based workflows. Particular attention is paid to deployment challenges, including system reliability, latency, cost optimization, and model behavior control. Industry engagement ensures exposure to real deployment environments, where participants examine how generative AI systems are integrated into enterprise workflows and decision-making processes.

Outcome: Participants will be able to design, deploy, and manage generative AI systems that deliver measurable operational and business value.

Certificate in Quantum Computing: Algorithms and Applications

This program focuses on the practical realization of quantum computing, bridging theoretical models and near-term applications. Participants implement key algorithms such as QAOA, VQE, and amplitude encoding, with a strong emphasis on hybrid quantum-classical systems. Hands-on work includes programming with leading quantum frameworks, simulation, and benchmarking. Participants ar also exposed to hardware constraints, including noise, scalability, and cost, through engagement with quantum labs and industry providers.

Outcome: Participants will be able to prototype and evaluate quantum algorithms, with a clear understanding of their applicability and limitations in real-world settings.

Certificate in Intelligent Decision-Making Systems

This program is centered on combinatorial optimization and decision-making in complex systems, addressing large-scale, multi-period, and often multi-objective problems that a in real-world operations. Participants work on problems characterized by: High dimensionality and combinatorial complexity; Dynamic, multi-stage decision processes; Multiple competing objectives (e.g., cost, efficiency, risk, sustainability) The program integrates optimization, reinforcement learning, and simulation, enabling participants to design systems that operate robustly under uncertainty. A key emphasis is placed on interpretability and reliability, ensuring that solutions are not only effective but also transparent, explainable, and safe for deployment in critical environments. Applications span domains such as logistics, healthcare systems, energy networks, and transportation, where decision-making must account for both operational constraints and systemic interactions.

Outcome: Participants will be able to design scalable, explainable, and robust decision systems for complex, real-world environments.

Certificate in Algorithmic Trading Strategies

This program focuses on advanced trading systems, with a particular emphasis on high-frequency trading, execution strategies, and AI-driven alpha discovery. Participants deve trading frameworks that integrate: Market microstructure analysis; Execution optimization and latency considerations; Portfolio construction and risk management. Both classical quantitative methods and modern approaches—such as machine learning and reinforcement learning—are applied to identify and adapt trading strategies. The program covers a broad range of markets and instruments, including: Traditional equities and derivatives; Cross-market and cross-asset strategies; Emerging domains such as cryptocurrencies, digital assets, and NFTs. Participants gain exposure to real-world trading environments, including infrastructure constraints, execution challenges, and regulatory considerations.

Outcome: Participants will be able to design, implement, and adapt trading strategies across markets, with a deep understanding of execution dynamics and system-level constraints.

Certificate in Decentralized Finance (DeFi)

This program focuses on the design and implementation of decentralized financial systems, combining blockchain infrastructure with economic and computational mechanisms. Participants engage in: Smart contract development and deployment; Design and issuance of digital tokens; Construction of decentralized applications (e.g., exchanges, lending platforms). A key component is the ability to design token economies, including incentive mechanisms, governance structures, and sustainability considerations. Participants also conduct security analysis, identifying vulnerabilities and systemic risks within smart contracts and protocols.. The program further addresses regulatory frameworks and ecosyste dynamics, providing a comprehensive understanding of both innovation and risk.

Outcome: Participants will be able to design, implement, and evaluate decentralized financial systems, including issuing and managing their own token-based platforms.

Certificate in Sustainable Intelligent Systems and Climate Finance

This program focuses on applying AI, optimization, and data-driven methods to sustainability and climate-related challenges. Participants work on large-scale systems such as: Energy networks and resource allocation; Carbon markets and emissions modelling; ESG analytics and sustainable investment strategies. The program emphasizes the integration of economic, environmental, and operational objectives, reflecting the inherently multi-objective nature of sustainability problems.

Outcome: Participants will be able to develop intelligent systems that balance economic performance with environmental impact, contributing to sustainable development.

Integrated Hackathon as a Core Learning and Talent Pipeline Mechanism

Global exposure (multi-country & multi-institution program)

Hands-on competence

Industry engagement